

WEEK 2 & 3: ORGAN SYSTEM

FUNCTIONAL ORGANIZATION OF NERVOUS SYSTEM

FUNCTIONS OF THE NERVOUS SYSTEM

1.) Receiving sensory input

- Monitors internal and external stimuli: 5 senses and pain, body position, temperature, blood pH, blood gases, and blood pressure

2.) Integrating information

- *Brain and spinal cord*
- May produce an immediate response, be stored as memory, or ignored

3.) Motor Output

- responds by activating effectors
- Skeletal muscles only contract *stimulated by nervous system* - controls major movements
- Also controls smooth and cardiac muscles, and many glands

4.) Maintaining homeostasis

- Nervous system detect, interpret and respond to variables

5.) Establishing and maintaining mental activity

- *Brain* (center for mental activity) -> consciousness, emotions, memory, and thinking

DIVISION OF THE NERVOUS SYSTEM

Central Nervous System

- Alters body activities through the information carried by PNS
- *Brain and spinal cord*

Peripheral Nervous System

- Links the CNS with the various parts of the body (carries information)
- *Cranial nerves, spinal nerves, ganglia*

- Sensory division or afferent (toward)
 - Conducts action potentials from the sensory receptors *to the CNS*
 - *Sensory neurons* - transmit action potentials from the periphery to the CNS
- Motor division or efferent (away)

- Conducts action potentials from the *CNS to effector organs*, such as muscle glands
- *Motor neurons* - transmit action potentials from the CNS toward the periphery
 - Somatic nervous system - CNS to skeletal muscles (voluntary)
 - Autonomic nervous system - CNS to cardiac and smooth muscle, and glands (involuntary) -> breathing, heart beat
 - **Sympathetic**
 - **Parasympathetic**
 - Enteric nervous system - has both sensory and motor neurons, can function without CNS. Regulates GI tract
- Interneurons (association with neuron, *within CNS*)

CELLS OF THE NERVOUS TISSUE

Neurons

- Receive stimuli, conduct action potentials, and transmit signals to other neurons or effector organ

Glial Cells (Neuroglia - nerve glue)

- Supportive cells of the CNS and PNS - do not conduct action potentials
- More plenty than neurons

NEURON STRUCTURE

Cell body (Soma)

- Single nucleus - source of information for gene expression

- Surrounded by *RER (nissl bodies)*, *golgi apparatus*, and *mitochondria*
- *Cytoplasm* - organized by plenty of neurofilaments (intermediate filaments) and microtubules

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- Decides

Dendrites

- Short, often highly branching cytoplasmic extensions that are tapered from their bases at the cell body to their tips
- Receives information from other neurons or other sensory receptors and transmit the information toward the cell body

Axon

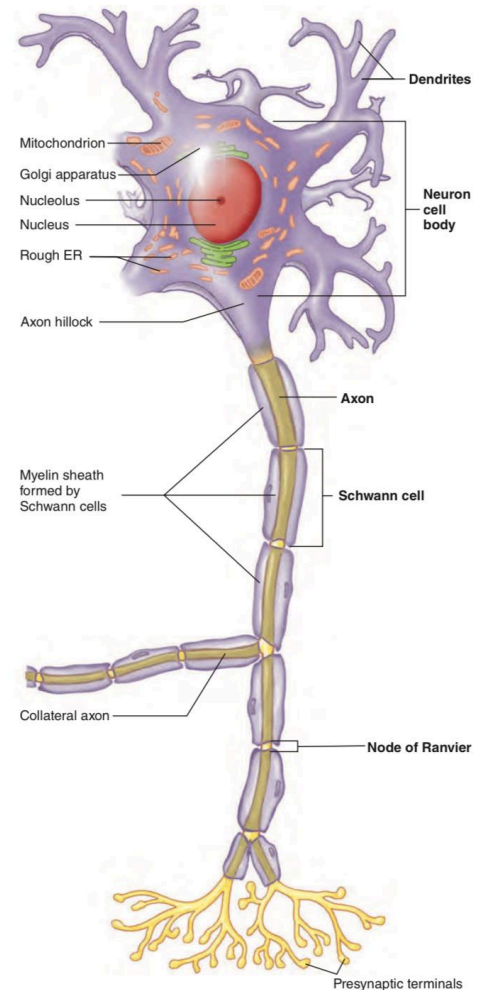
- Single long cell process extending from the cell body
- Has consistent thickness, but differs in length (few mL to more than M)
- Conducts action potentials towards the CNS (sensory neurons) and conduct action potentials away from the CNS (motor neurons)
- Can be surrounded by a highly specialized insulating layer of cells called the myelin sheath
- Sends information to axon terminal

Axon hillock

- Area where the axon leaves the cell body, and where AP starts
- In between cell body and axon

Axon Terminal (Synaptic knob)

- Where neurotransmitters are stored



TYPES OF NEURONS

Multipolar neurons

- Many dendrites and one axon
- Found within CNS and *most of* motor neurons

Bipolar neurons

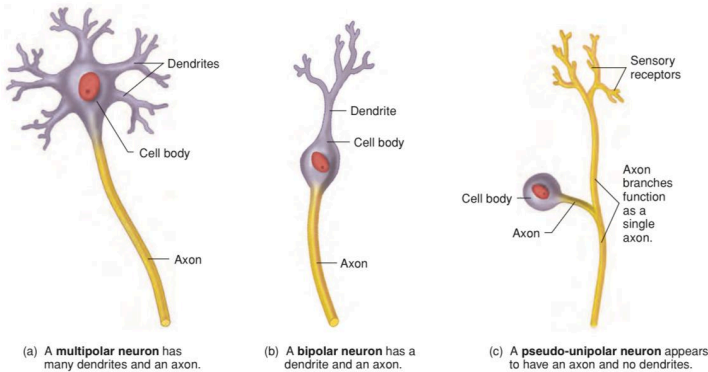
- Two processes: one dendrite and one axon
- Located in some sensory organs (*retina* and *olfactory epithelium*)

Unipolar neurons

- Single process extending from the cell body
- Sensory neurons in dorsal root ganglia
 - Divides into two processes after leaving the cell body (functions as a single axon)
 - *Toward the body's periphery*
 - *Toward the CNS*

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- Received sensory information at the periphery and transmit it through action potentials to the CNS



- Forms myelin sheath only in one axon segment in PNS
- Might help in repair by forming regeneration tube that guides the regrowth of the axon after injury
- High regeneration

Satellite Cells

- Surround and support neuron cell bodies in the ganglia (cluster of neuron cell bodies in the PNS)
- Provides nutrients and remove wastes
- Help maintain a stable chemical environment

NERVOUS TISSUE RESPONSE TO INJURY CNS

- *Limited regeneration* (due to inhibitory environment, lack of neurolemma - protective covering)
- Stroke, SCI, TBI

PNS

- *Potential regeneration* if soma is intact and Schwann cells for regeneration tube
- **Glial scar formation** - astrocytes wall off injury site (no nutrients and blood supply)
- Saturday night palsy, carpal tunnel syndrome
- **Neuroplasticity** - remaining neurons form new connections to compensate for lost function (found more in PNS)

MYELINATED V.S. UNMYELINATED AXONS

Myelin sheaths

- Specialized layer that wrap around the axons of some neurons
- Excellent insulator that prevents most of ion movements across the membrane
- **Nodes of Ranvier** - gap in between (mm)
 - Ion movement can occur

Myelinated - motor

- Increases the speed and efficiency of action potentials propagation along axon

GLIAL CELLS OF CNS

Astrocytes

- Major supporting cell in the CNS
- Can stimulate or inhibit the signaling activity of nearby neurons
- Participate with the blood vessel endothelium to BBB
- Helps limit damage to neural tissue - forms a scar for the degeneration of damaged axons

Ependymal cells

- Line the fluid-filled cavities (ventricle and canals) within the CNS
- Produces CSF

Microglia

- Act as immune cells of the CNS - phagocytosis
- Help brain by removing bacteria and cell debris

Oligodendrocytes

- Forms myelin sheath in multiple axon segments in CNS
- Weak regeneration

GLIAL CELLS OF PNS

Schwann Cells

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- Formed by the cell processes of oligodendrocytes (CNS) and Schwann cells (PNS)

- Repeatedly wraps around a segment of an axon to form a series of tightly-wrapped cell membranes

Unmyelinated

- Rest in indentations of the oligodendrocytes or Schwann cells
- Usually found in axons of multiple neurons
- Continuous and slower

ORGANIZATION OF NERVOUS TISSUE

Gray Matter

- Consists of groups of neuron cell bodies, dendrites, and unmyelinated axon: outermost
- Rich with blood supply - slower signal
- Process and analyze information
 - Cortex - In CNS
 - Nuclei - Deeper gray matter
 - Ganglion - Cluster of neuron cell body in PNS

White matter

- Consists of bundles of myelinated axons and oligodendrocytes: innermost
- Communication highway - faster signal
 - Nerve tract (conduction pathway) - propagate action potentials from one area of CNS to another
 - Nerves - Bundles of axons and associated connective tissue in PNS

PERMEABILITY CHARACTERISTICS OF PLASMA MEMBRANE

- Ions cannot move freely to the plasma membrane. Thus needs to flow through ion channels (protein embedded)
- **Two basic types of ion channels**
 - Leak channels (always open) - MP is established by diffusion of ions through leak channels

- Gated channels (closed until opened by specific signals) - can change MP and is responsible for the action potential
 - Chemically-gated channels - opened by neurotransmitters
 - Voltage-gated channels - opened by change in MP

ESTABLISHING THE RESTING MEMBRANE POTENTIAL

- **Polarized** - the electrical gradient of the cell
- **Resting membrane potential** - uneven charge distribution when the cell is at rest, or results from the difference of ion concentration across the membrane
- Factors why it's generated:
 - Higher concentration of K inside the cell
 - Higher concentration of Na outside the cell
 - Greater permeability of the cell membrane to K than to Na
- K leak channels > Na leak channels
 - Resting membrane potential has greater permeability to K (K have greater contribution to RMP)

In resting state, high K outside and high Na inside
-> K and Na leak channels are open -> K goes in and Na goes out (they also try to go opposite ways but mainly stopped by the ion and membrane charge) -> Na/K pump pumps 3 Na out and 2 K in to maintain balance -> once electrical equilibrium is established, so does the RMP

CHANGING THE RMP

- **Resting membrane potential**: Na channels and most, but not all, K channels are closed (outside is positive; inside is negative)
- **Depolarization**: Na channels open and K channels begin to open. Results because of the inward movement of Na, which makes the inside of the membrane positive

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- **Repolarization:** Na channels open and additional K channels open. Na movement into the cell stops, and K movement out of the cell increases causing repolarization

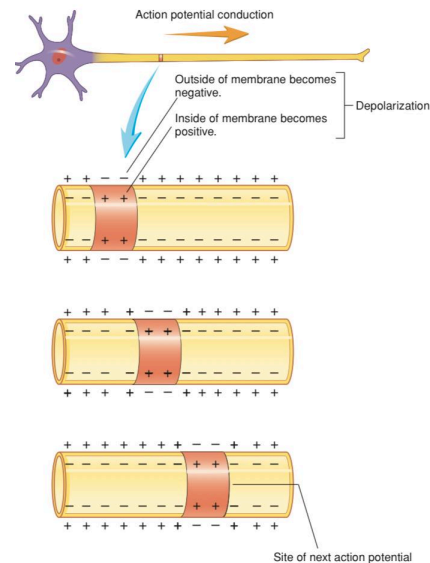
ACTION POTENTIALS

- **Local current** - when stimulus (neurotransmitter) is applied to muscle or nerve cell, chemically gated channel opens, opening Na channels briefly

- Movement of Na inside is called *local current* - causes the inside of the cell to become positive (depolarization) -> cause local or graded potential
- If depolarization is not strong enough, channels close again. Thus, local potential disappears
- If depolarization is strong enough. Na enters the cell so that local potential reaches threshold
 - Causes voltage-gated Na channels to open
 - Often reached in the axon hillock
 - Opening of channels causes 600-fold increase in membrane permeability to Na
 - Voltage-gated K channels begin open
 - Na continues to enter until the inside of the membrane is positive as opposed to the outside
 - Causes Na channels to close, and K channels to open
 - Na stops entering the membrane, and K leaves (K permeability only lasts for a very short time)
 - This causes repolarization the cell membrane to its RMP
 - Hyperpolarization - At the end of repolarization, the charge of the membrane briefly becomes more positive than RMP

- Depolarization and repolarization constitute an AP

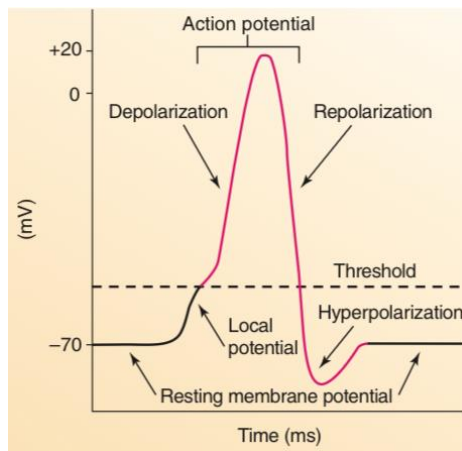
- Inside of membrane is negative, and outside is positive -> once AP was propagated along the membrane, it releases local current ONLY adjacent to the AP and depolarizes (making inside +; outside -) it -> when depolarization reaches threshold, a new action potential is made adjacent/next to it -> once the AP became absent, it causes repolarization -> repeat



- **All-or-none fashion** - if threshold is reached, action potential occurs and vice versa at axon hillock
- Action potentials are all of the same magnitude and size. The neurotransmitter or the stimuli don't change the size, just the frequency of the action potentials
 - Thus, the strength of the nervous system is determined by the number of times the action potential occurs
- Conducted unmyelinated (slowly) and myelinated (faster) axons
 - **Continuous conduction** - happens step-by-step in unmyelinated axons
 - **Saltatory conduction** - action potential jumps from one node of Ranvier to another (nodes of Ranvier makes the entire membrane unnecessary)

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Basically, RMP starts first (-70mV). Then if the stimulus (neurotransmitter) is strong enough, it will reach a threshold, creating GP. Once GP is established, Na channels open and Na enters the membrane making it more positive (depolarization -55mV). AP is made at this point. Afterwards, Na channels close and K channels open, releasing K outside (repolarization). K channels do not close immediately. Instead, it becomes more negative than the RMP (hyperpolarization - more than -55mV). K channels will close after that, and the Na/K pump will establish the resting state again. Action potential might undergo refractory period (either absolute or relative)



REFRACTORY PERIOD

- The time after a neuron fires action potentials and became either less responsive or unresponsive to
- Make sure action potentials don't pile up
- Prevents seizure or overstimulation, and backward signalling

- Absolute refractory period

- No new action potential can be created no matter how strong the stimulus is
 - Na channels are inactivated
- Ensures action potential is unidirectional

- Relative refractory period

- Following the refractory period, neuron can fire again, but only if the stimulus is stronger than usual

- Helps control the frequency of action potentials
 - K channels are open which causes the membrane to be hyperpolarized. Thus, making it harder for it to reach threshold

SYNAPSE

- a junction where the axon of one neuron interacts with another neuron or with cells of an effector organ (muscle or gland)

- **Presynaptic terminal** - end of an axon
 - Neurotransmitters - chemical substances stored in synaptic vesicles in the presynaptic terminal
- **Postsynaptic terminal** - membrane of the dendrite or effector cell
- **Synaptic cleft** - space separating the 2 terminals

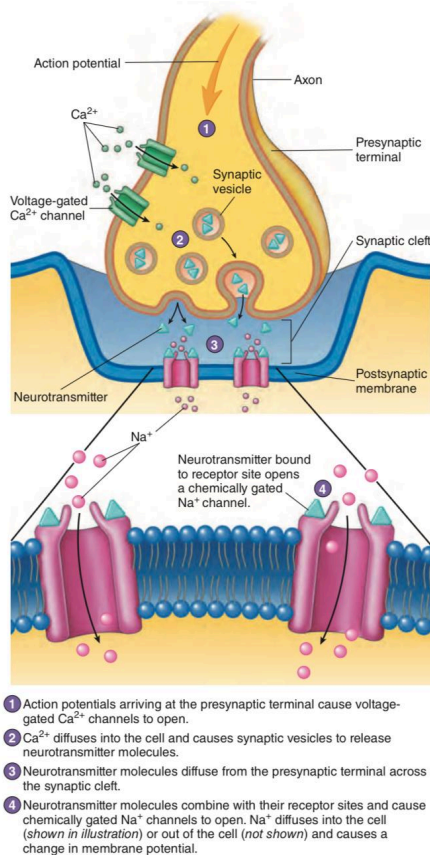
- Action potential reaches the presynaptic terminal, and voltage-gated Ca channels open releasing Ca into the presynaptic terminal

- Influx of Ca causes the release of neurotransmitters which diffuses across synaptic cleft and bind to specific receptor molecules on the postsynaptic membrane
- This causes chemically gated channels (Na, K, Cl) to open or close in the postsynaptic membrane
 - Depends on the neurotransmitters in the presynaptic membrane, and receptors in the postsynaptic membrane

- It can either stimulate or inhibit

- E.g. If Na channels open, postsynaptic will be depolarized, which makes it ready for threshold (action potential will be created)
- E.g. If K or Cl channels open, the K and Cl will naturally move out of the cell (postsynaptic will be negative) therefore making the inside more negative

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ELECTRICAL SYNAPSES

Neurons are connected by gap junction (special channels for ions and small molecules) -> once action potential reaches presynaptic terminal, it can directly pass through the gap junction to the postsynaptic terminal -> signal travels from one neuron to the next without neurotransmitters

CHEMICAL SYNAPSES

Action potential reaches presynaptic terminal and voltage-gated Ca channels open to release Ca -> Ca is absorbed by the presynaptic terminal -> releases neurotransmitter from vesicle to the synaptic cleft -> neurotransmitter binds to postsynaptic terminal -> it can either excite or inhibit the terminal depending on the neurotransmitter -> opens chemically gated channels to open -> trigger action potential -> neurotransmitter is removed (either reused in

presynaptic terminal or broken down by enzymes in synaptic cleft)

OVERVIEW OF NEUROTRANSMITTERS

- Rapidly broken down by enzymes within the synaptic cleft or are transported back to the presynaptic terminal
- Removed from the synaptic cleft therefore the effects are very short-term
- Can be released or broken down easily that postsynaptic cell can be stimulated many times a second

• Acetylcholine

- Acetylcholinesterase - breaks down ACh
 - Products are returned to presynaptic terminal for reuse
- **Site of Release:** CNS, ANS, and neuromuscular junction synapses
- **Effect:** both excites and inhibits
- **Clinical example:**
 - *Alzheimer disease* (senile dementia) - decrease in ACh-secreting neurons
 - *Myasthenia* (weakness of skeletal muscles) - reduced ACh receptors

• Norepinephrine

- Can either actively transported back into the presynaptic terminal or broken down by enzymes
- **Site of Release:** selected CNS and some ANS synapses
- **Effect:** excites
- **Clinical Example:**
 - *Cocaine* (blocks reuptake) and *amphetamine* (increases and blocks reuptake) causing in overstimulation of postsynaptic neurons

• Serotonin

- **Site of Release:** CNS synapses

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- **Effect:** generally inhibits
- **Clinical Example:**
 - Mood, anxiety, and sleep induction
 - *Schizophrenia* - elevated
 - *Depression and anxiety disorders* - Prozac blocks serotonin transporters
- **Dopamine**
 - **Site of Release:** Selected CNS and some ANS synapses
 - **Effect:** both excites and inhibits
 - **Clinical Example:**
 - *Parkinson disease* (depression of voluntary motor control) - destruction of dopamine-secreting neurons
- **Gamma-aminobutyric acid (GABA)** - benzodiazepine (XANAX), barbiturates - sedative, baclofen - muscle relaxant, zolpidem - insomnia
 - **Site of Release:** CNS synapses
 - **Effect:** Inhibits
 - **Clinical Example:** *Epilepsy* (excessive discharge of neurons) - low in GABA
- **Glycine**
 - **Site of Release:** CNS synapses
 - **Effect:** Inhibitory
 - **Clinical Example:** *Strychnine* - blocks glycine receptors, which causes muscle contraction
- **Endorphin**
 - **Site of Release:** descending pain pathways
 - **Effect:** Inhibitory
 - **Clinical Release:** drugs like *opiates morphine* and *heroin* bind on presynaptic neurons - reduces pain by blocking the release of endorphin

- **Summation** - many presynaptic action potentials are combined in order to reach threshold

- **Spatial summation**

- Local potentials came originate from different locations on the postsynaptic neuron (e.g., converging pathway)

- **Temporal summation**

- Local potentials overlap in time (came from only one that fires rapidly)

NEURONAL PATHWAYS AND CIRCUITS

Simplest pathway

1.) *Converging pathway*

- Two or more neurons synapse with the same neuron
- More than one neuronal pathway converge into one

2.) *Diverging pathway*

- One neuron divides and synapses with more than one other neuron
- Information in one neuronal pathway to diverge into two or more

SPATIAL AND TEMPORAL SUMMATION

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FUNCTIONAL NEUROANATOMY

MAJOR DIVISIONS

1.) CNS - receives information, control

- Brain and spinal cord

2.) PNS - delivers information, pathway

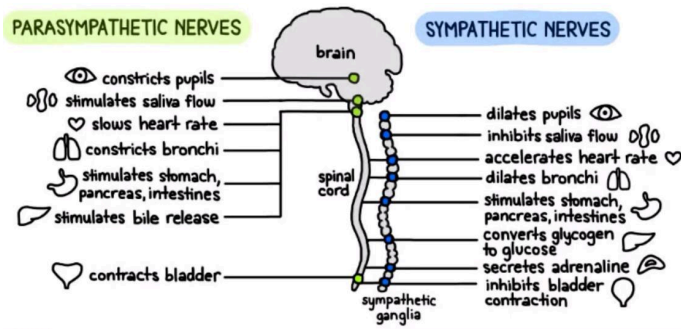
- SNS - voluntary movements
- ANS - involuntary movements
 - Sympathetic nervous
 - Parasympathetic nervous
- Enteric

- Cranial nerves

- Spinal nerves

- Ganglia

- Peripheral receptor



BRAIN

MAJOR PARTS

1.) Cerebrum

- Largest part, two hemisphere, divided by longitudinal fissure

- Functions: higher thought, voluntary movement, sensory processing, memory, emotions

- Lobes:

- *Frontal* - speech (Brocha's area), planning, personality, motor control
- *Parietal* - sensory perception, spatial awareness
- *Temporal* - auditory, memory, language (Wernicke's area)
- *Occipital*: vision processing

2.) Diencephalon

- Thalamus - sensory relay center

- Hypothalamus - autonomic control, endocrine link, homeostasis

- Epithalamus - includes pineal gland (melatonin, sleep-wake cycle or circadian rhythm)

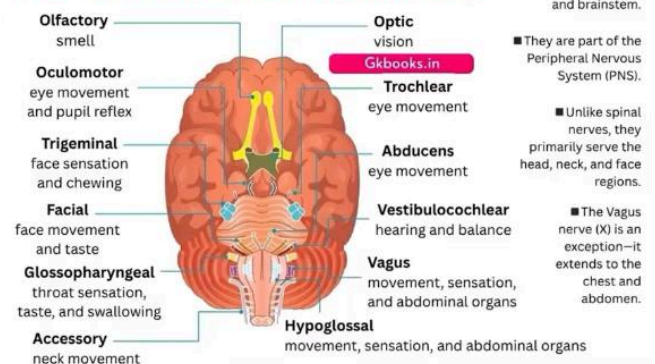
3.) Brainstem - bridge that connects brain and spinal cord

- Midbrain - visual/auditory reflexes, superior cerebellar peduncles decussate

- Pons (bridge) - relay between cerebrum and cerebellum; respiratory control

- Medulla oblongata - vital centers (respiration, HR, BP); reflexes (cough, swallow, sneeze); where upper motor neuron fibers of the corticospinal tract decussate

12 CRANIAL NERVES



No.	Cranial Nerve Name	Type	Function (Key Points)
I	Olfactory	Sensory	Smell
II	Optic	Sensory	Vision
III	Oculomotor	Motor	Eye movement, pupil constriction, eyelid elevation
IV	Trochlear	Motor	Eye movement (superior oblique muscle)
V	Trigeminal	Both	Facial sensation, chewing
VI	Abducens	Motor	Eye movement (lateral rectus muscle)
VII	Facial	Both	Facial expressions, taste (anterior 2/3 tongue), tears
VIII	Vestibulocochlear	Sensory	Hearing and balance
IX	Glossopharyngeal	Both	Taste (posterior 1/3), swallowing, salivation
X	Vagus	Both	Parasympathetic control of heart, lungs, digestive tract
XI	Accessory (Spinal)	Motor	Shoulder and neck movement (trapezius, sternocleidomastoid)
XII	Hypoglossal	Motor	Tongue movement

EMO3, SO4, LR6

Vagus - longest

Trigeminal - largest

CN 1 - anosmia

CN 3 - muscle opening the eye; located at the midbrain at the level of superior colliculus; berry aneurysm (PCA & ICA); ptosis, mydriasis, and the deviated "down and out"

CN 4 - only CN that emerges from the dorsal aspect of brainstem, stupid nerve

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CN 5 - carries afferent limb of the corneal reflex (facial sensation)

CN 7 - muscle closing the eye

CN 9 - parotid gland

4.) Cerebellum "little brain" - fixes the movement

- At the back, has two hemisphere: anterior and posterior lobe, divided by primary fissure
- Coordination, posture, balance (vestibular system), motor learning
- Apraxia - shaky, uncoordinated movement

5.) Basal Ganglia - found in the innermost part of cerebrum, beneath the cortex

- Voluntary initiation control of movement
 - Caudate nucleus - C-shaped structure for motor processing, learning, and goal-directed behavior
 - Putamen - works with caudate for motor function
 - Globus pallidus
 - External segment - modulates subthalamic nucleus activities
 - Internal segment - main output to thalamus, influencing motor activity
 - Subthalamic nucleus - regulates movement through excitatory input
 - Substantia nigra
 - Pars compacta - produces dopamine for movement initiation
 - Pars reticulata - works with GPI in motor output
 - Nucleus accumbens - reward, motivation, and reinforcement learning (limbic circuit)
- Parkinson's disease - *nakakuba*

PROTECTIVE LAYER OF THE BRAIN

1.) Meninges

- Dura mater (outer, toughest)
- Arachnoid matter (middle, CSF circulation)
- Pia mater (inner, adheres to the brain surface)

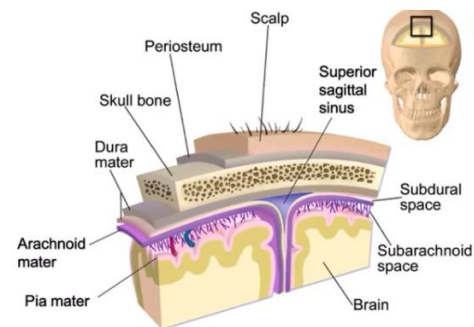
2.) Cerebrospinal fluid - shock absorber

- Produced ependymal cells, and first stored in choroid plexus in ventricles
- Functions: cushions, buoyancy, nutrient delivery, waste removal

Choroid plexus -> lateral ventricle -> interventricular foramen (foramen of **Monro**) -> third ventricle -> **Sylvian** aqueduct -> fourth ventricle -> median aperture (foramen of **Magendie**) -> lateral aperture (foramen of **Luschka**) -> subarachnoid space -> arachnoid villi

3.) Blood-brain barrier - body guard; filters

- Tight endothelial junctions + astrocyte support
- Prevents entry of toxins, maintains stable environment



Skin, Connective tissues, Aponeurosis, Loose areolar connective tissue, Periosteum

SPINAL CORD

- Extends from the medulla oblongata -> ends between L1-L2 vertebra level
- Protected by vertebrae, meninges, CSF
- Gray matter (butterfly-shaped, H-shaped)
 - Dorsal horn - sensory
 - Ventral horn - motor
 - Lateral horn - autonomic (T1-L2)
- White matter
 - Ascending tract (sensory -> brain)
 - Descending tract (motor -> periphery)
- Spinal nerves
 - 31 pairs = 62 in total
 - 8 cervical, 12 thoracic, 5 lumbar, 5 sacral, 1 coccygeal

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ENTERIC NERVOUS SYSTEM

- Second brain of the gut
- Network in GI tract wall
- Regulates motility, secretion, blood flow independently of CNS, but influenced by ANS

SENSORY RECEPTORS

- Mechanoreceptors - touch, pressure, vibration
- Thermoreceptors - temperature; (Krause-end bulbs)
- Nociceptors - pain
- Photoreceptors - vision
 - Rod - dark
 - Cone - light
- Chemoreceptors - smell, taste blood chemistry

MECHANORECEPTORS

- 1.) Pacinian corpuscles** - vibration & deep pressure
 - Located in the deep layers of the skin, joints, and ligaments
 - Detect rapid vibrations or strong pressure
- 2.) Meissner's corpuscles** - light touch and texture
 - Found in the papillary layer of the skin, especially in fingertips, lips, and palms
 - Sensitive to gentle touch, texture and fluttering
 - Feeling the texture of rough or smooth objects
- 3.) Merkel discs** - pressure and texture
 - Found in the epidermis, especially in the fingertips
 - Detect steady pressure and fine details of texture
 - Holding a pen, feeling a texture
- 4.) Ruffini endings** - stretch and skin tension
 - Found in the dermis, joints, and ligaments
 - Detect skin stretch and sustained pressure
 - Holding door knob while opening the door
- 5.) Hair follicle receptors** - hair movement
 - Wrap around hair follicles
 - Detect light touch and hair movement

Dorsal column-lemniscus pathway - fine touch, vibration, and proprioception

Lateral spinothalamic - pain and temperature

- decussation happens one or two segments above the entry of the sensory fibers

BASIC EMBRYOLOGY

CNS PATTERNING

1.) Ectoderm

- Nervous system, hair, skin
 - Neurulation - neural plate folds and fuses to form neural tube (inside; CNS), and neural crest (outside; PNS)
 - Neural tube - 22nd-23rd day; Anterior neuropore - 25th day; Posterior neuropore - 27th day
 - **Failure:** Anencephaly (cranial defect), spina bifida (spinal defect)
 - Deficiency in folic acid
 - MTHFR gene defect
 - Maternal disease (diabetes, obesity)
 - Anti-epileptic drug, drinking, and smoking during pregnancy

2.) Mesoderm

- Structural framework
- Muscle, skeleton, kidneys, reproductive system, blood vessels, heart

3.) Endoderm

- Lines internal organs
- Endocrine system, lungs, digestive tract, liver

CNS PATTERNING AXES

1.) Anterior-posterior axis (head-to-tail)

- Cranial end: develops primary brain vesicle
 - **Forebrain** - prosencephalon -> *telencephalon* (cerebrum, basal ganglia) & *diencephalon* (thalamus, hypothalamus, epithalamus)
 - **Midbrain** - mesencephalon -> *mesencephalon* (midbrain in brainstem)
 - **Hindbrain** - rhombencephalon -> *metencephalon* (cerebellum and

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pons) & *myelencephalon* (medulla oblongata)

- Caudal end: develops into the spinal cord

2.) Dorsal-ventral axis (back-to-belly)

- Defined by chemical signaling molecules - morphogens

- **Ventralizing signal**:

- Sonic hedgehog (Shh) - secreted by the notochord and floor plate
- Includes vertebral structure such as:
 - *Motor neurons (ventral horn)*

- **Dorsalizing signal**:

- Bone morphogenic proteins (BMPs) - secreted by the ectoderm and roof plate
- Induces dorsal structure such as:
 - *Sensory neurons (dorsal horn)*

- **Intermediate zone** - combination of dorsalizing and ventralizing

- interneurons that connect sensory and motor pathways (important for reflexes)

ADULT BRAIN

Gyri, Sulci, and Fissures

- **Gyrus** - raised ridge or fold of the cerebral cortex, increase surface area of cerebral cortex = more neurons

- Precentral (front of central sulcus) - primary motor cortex (**BA 4**)
- Postcentral (behind central sulcus) - primary somatosensory cortex (**BA 3, 1, 2**)
- Superior temporal - auditory cortex (**BA 41 & 42**)
- Cingulate - part of the limbic system, learning, memory
- Fusiform - face and word recognition (**BA 37**)

- **Sulcus** - shallow groove between gyri

- Central (Rolandic) - separates frontal (motor) and parietal (sensory) lobe
 - Between precentral (motor) and postcentral (sensory) gyrus

- Parieto-occipital - separates parietal and occipital lobe

- Calcarine - found in the occipital lobe, contains primary visual cortex (**BA 17**)

- **Fissure** - deep, prominent groove (larger than sulci) that separates major brain regions and lobes

- Longitudinal - separates left and right hemisphere
- Transverse - separates cerebrum and cerebellum
- Lateral (Sylvian) - separates temporal to frontal and parietal lobes

BRODMANN'S AREAS

- Defined by German neurologist, **Korbinian Brodmann** (1909)

- Regions of cerebral cortex

- Identified **52 distinct areas** which correspond to specific motor, sensory, and cognitive function

- **Arcuate fasciculus** - connects Wernicke's and Broca's

- Lesion: can't speak and understand

FRONTAL LOBE - motor

4 - Primary Motor Cortex

- Lesion: C/L hemiplegia/paralysis

6 - Premotor Cortex

- Initiation, planning, coordination
- Lesion: Apraxia - difficulty in performing movement when verbally instructed
 - Ideomotor - can't do task without the presence of true weakness
 - Ideational - does different task

8 - Frontal/eyefield

- Conjugate movement of the eyes
- Lesion: impaired gaze

9-12 - Prefrontal

- Judgment, intelligence, behavior,

44 - Broca's

- Speech motor area; speaking
- Lesion: non-fluent aphasia

PARIETAL LOBE - sensory

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3, 1, 2 - Primary Somaesthetic Cortex

- Lesion: C/L hemianesthesia/sensory loss

5, 7 - Somastethic Association Area

- Barognosis - weigh; grapesthesia - writing in body; stereognosis - object's shape; proprioception - spatial recognition; kinesthesia - movement difference
- Lesion: neglect syndrome

39 - Angular Gyrus

- Lesion: Gertsman syndrome

40 - Supramarginal Gyrus

- Lesion: ideomotor apraxia - can't do task without the presence of true weakness

43 - Primary gustatory area

- Taste sensation

TEMPORAL LOBE - hearing

41 - Primary Auditory Cortex

- Pitch, loudness, tone, rhythm
- Lesion if one: Partial C/L deafness
- Lesion if two: Cortical deafness

42 - Secondary Auditory Cortex / Auditory Association Area

- Recognizing music and speech
- Dyslexia

22 - Auditory Association Area

- Left temporal lobe - Wernicke's Area - comprehension
- Right temporal lobe - auditory agnosia

OCCIPITAL LOBE - vision

17 - Primary Visual Area / Striate Cortex

- Lesion if one 17: Homonymous hemianopsia
 - Optic nerve - ipsilateral
 - Optic tract - contralateral; both eye with problems but with specific side
 - E.G., Lesion in right OT affects nasal of RE and temporal of LE: left homonymous hemianopsia
- Lesion if two 17: Cortical blindness / full blindness

18-19 - Visual Association Area / parastriate Cortex

- 18 - Color blindness / dyschromatopsia
- 18 & 19 - Visual agnosia - can see but can't process
 - Prosopagnosia - inability to recognize object despite having normal visual process
 - Anosognosia - inability to recognize and process faces

BLOOD SUPPLY OF THE BRAIN

CIRCLE OF WILLIS

- **Collateral circulation** - protective role

- Comes from **aortic** arc - largest artery

- **Polygonal anastomosis** - collateral circulation if a vessel is blocked

- If basilar artery is blocked, ICA from PCOM supplies PCA

- **MCA** - most common artery occluded

- **ACA** - most common site of aneurysm

Main branches:

1.) Brachiocephalic trunk

2.) Left subclavian artery

- R subclavian is not considered because it is a part of the brachiocephalic

3.) Left common carotid artery

- R common is not considered because it is a part of the brachiocephalic
- Supplied the left side of the head and neck

FLOW

Anterior Part

- From the aortic arc, it will branch out to two parts:

- Right external carotid artery and right internal carotid artery
- Left external carotid artery and left internal carotid artery

- **ICAs** will further ascend to the head/skull (C3-C4), and are subdivided into three branches:

- Middle cerebral artery (MCA) - frontal, lateral surface of brain, not part of circle of Willis, **largest artery** in anterior circulation, *stroke*

WEEK 2 & 3: ORGAN SYSTEM

- Anterior cerebral artery (ACA) - medial/frontal parietal cortex
 - *Parietal* - lateral
 - *Some frontal* - inferior
 - *Some temporal* - medial
- Posterior communicating artery (PCOM)
 - Connects internal carotid artery (anterior) and basilar artery (posterior)
 - Lesion: vision loss

- Anterior communicating artery (ACOM)

- Provides collateral flow between the left and right hemisphere if one ICA is occluded

Posterior Part

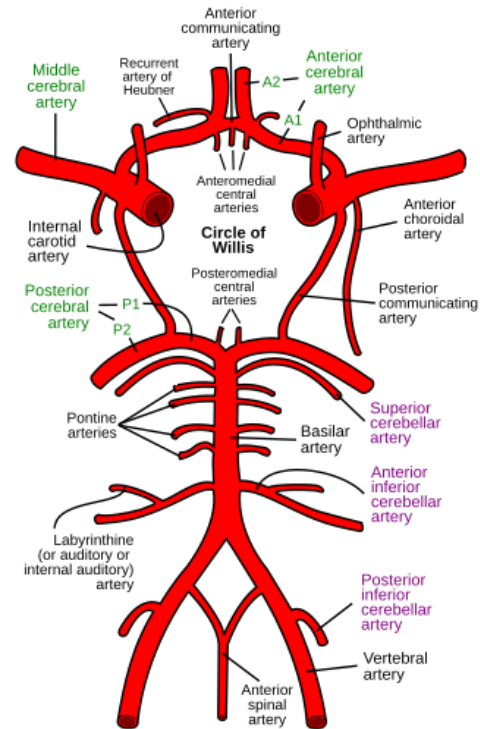
- From the left and right subclavian arteries will ascend to the neck (vertebral level) to form *vertebral artery*
- From the carotid canal of the vertebral artery, it will combine and form *basilar artery (largest artery in posterior circulation)*
- Basilar arteries will further subdivide to superior cerebellar artery
 - Posterior cerebral artery (PCA)
 - Terminal branch of basilar artery, *occipital lobe and inferotemporal lobe*, visual cortex

- **Pontine artery:** Pontine lesion - BA 4, 5, 6, 7, and upper part of cranial nerve number 8 may be affected. Collateral circulation prevents the defects to all BAs
- Also includes anterior inferior cerebellar artery, postero-inferior cerebellar artery
- Labyrinthine artery or internal auditory artery (inner ear)

Anastomosis (Circle of Willis)

- Anterior cerebral artery (ACA) connected by anterior communicating artery (ACOM)
- Internal carotid artery (ICA) is connected to posterior carotid artery (PCA) through posterior communicating artery (PCOM)
- Occlusion patterns:
 - MCA - C/L face/arm weakness; aphasia

- ACA - C/L leg weakness
- PCA - visual field defects
- ACOM - classic bitemporal hemianopsia
- VBA - coma/death
- If basilar is blocked, ICA and PCOM supplies PCA
- ICA connects to PCA by PCOM



HOMUNCULUS - "Little man"

- Visual representation of how different parts of the body are mapped in cerebral cortex
- Found in the primary motor cortex and primary somatosensory cortex
 - Anteromedial - ACA
 - Lateral - MCA
 - Posterior - PCA

TYPES:

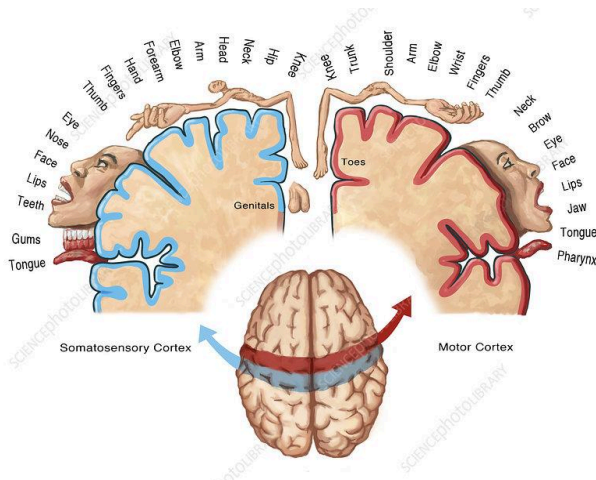
- Motor homunculus - controls voluntary movement
- Sensory homunculus - receives sensations

KEY RULE:

- The more sensitive or precise the body part is, the larger brain area is devoted for it

WEEK 2 & 3: ORGAN SYSTEM

- E.g., hands, lips, touch - large areas -> with complex skills
- E.g., trunk, legs, back - small areas -> supports movements



BLOOD SUPPLY IN RELATION TO HOMUNCULUS:

- Anteromedial: ACA (lower extremities, genitals)
- Lateral: MCA (face, upper extremities)

- Posterior: PCA (vision, vital organs)

Hoarseness and loss of gag reflex

- CN 9 and 10: basilar artery

Bitemporal hemianopsia

- ACOM

Cannot abduct eye

- CN 6

Lesion of CN 8 due to aneurysmal compression

- Anterior inferior cerebellar artery

Contralateral hemianopsia and alexia without agraphia

- PCA

VBA and can be affected in lateral medullary syndrome

- CN 9 & 10

Compression of CN 3 by a PCOM aneurysm typically first affects:

- Parasympathetic fibers to the pupil